GLOBAL ELECTRIC VEHICLE MARKET ANALYSIS

How EVs Are Transforming the Automobile Industry Across Countries, 2010-2025

Final Individual Project — Data Visualization, Summer 2026

60 countries · 16 years (2010-2025) · IEA Global EV Outlook 2025 data, via Our World in Data

The Dataset: A Decade and a Half of EV Adoption

One row per country per year, 2010-2025, built from four IEA Global EV Outlook 2025 series republished by Our World in Data: annual EV sales, EV sales as a share of all new car sales, EV stock (the installed fleet), and the battery-electric (BEV) share of EV sales. Enriched with continent groupings, an auto-manufacturing-powerhouse flag, market-phase tiers, and each country's 'tipping point' year.

60

Countries tracked

27

Crossed the 5% tipping point

7

Reached 50% EV sales share

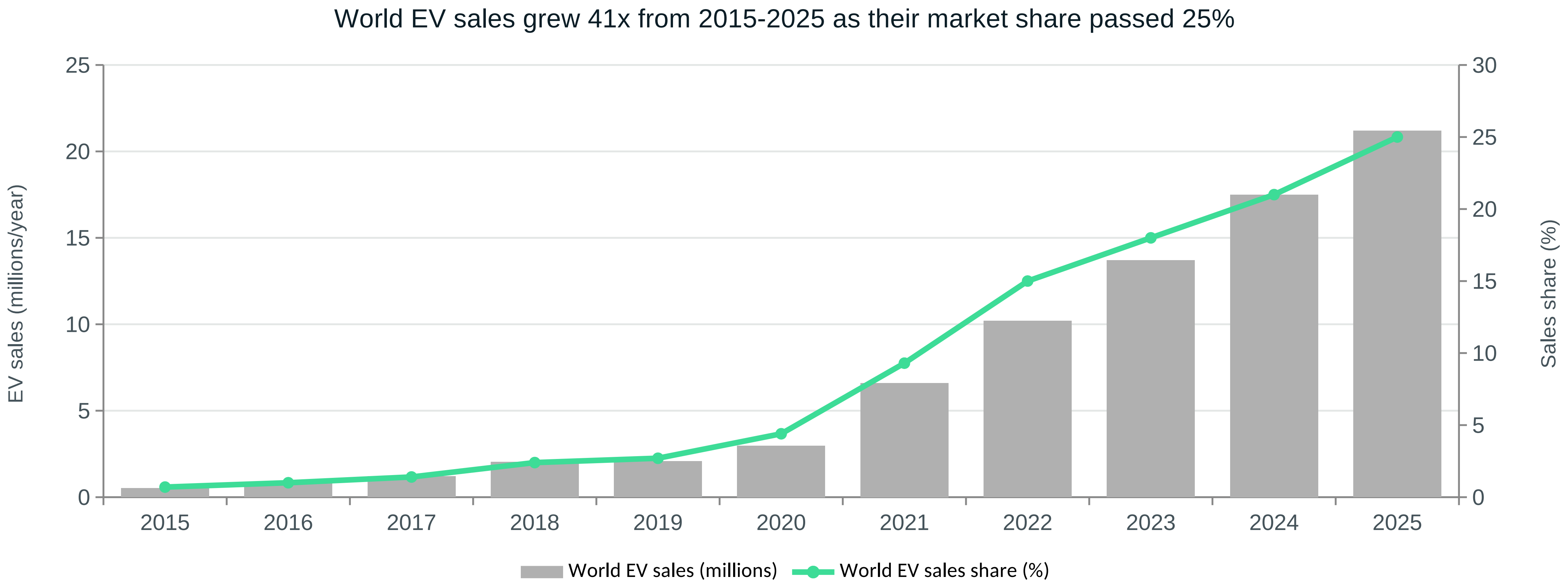
40.8x

World EV sales growth, 2015-25

Source: International Energy Agency, Global EV Outlook 2025 — https://www.iea.org/reports/global-ev-outlook-2025, via Our World in Data (https://ourworldindata.org/electric-car-sales).

'Auto-manufacturing powerhouse' is a fixed list of the world's largest light-vehicle producing nations (China, USA, Japan, Germany, India, South Korea, and others) used to compare adoption speed against markets with no legacy internal-combustion industry to protect — an analytical grouping, not a value judgement.

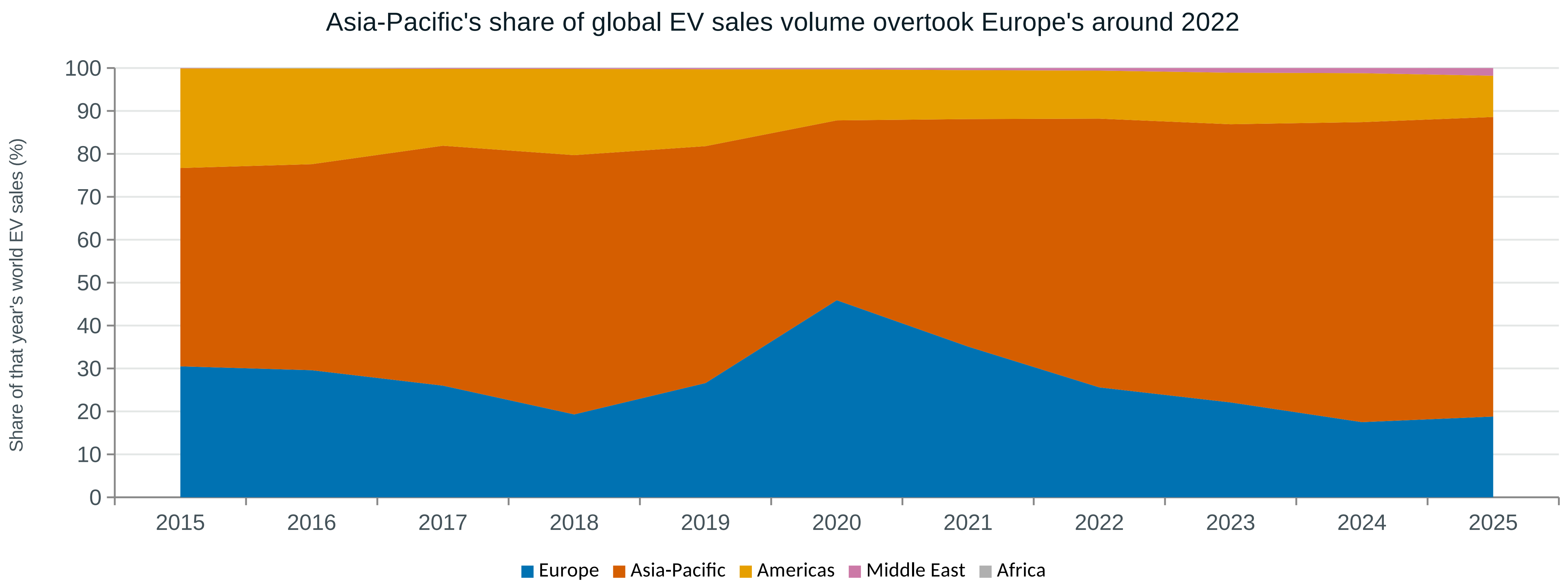
Q1 How fast has the world actually gone electric, and who is driving that volume?



Takeaway: China alone rose from 40.6% to 62.7% of that global sales volume -- the "global" EV transformation is, in raw units, disproportionately a Chinese one.

Q2

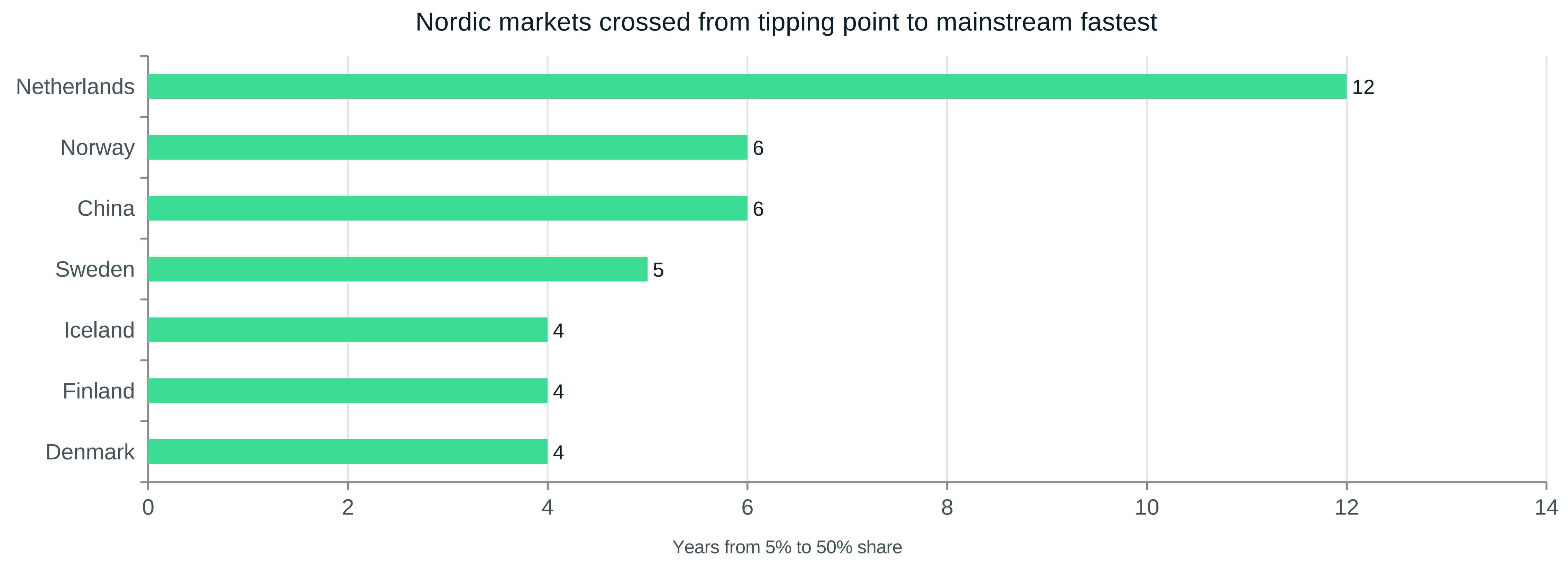
Which continent leads the world's EV transformation today?



Takeaway: Europe held the clearest early lead, but Asia-Pacific's volume (driven overwhelmingly by China) has commanded roughly two-thirds of world EV sales since 2022.

Q3

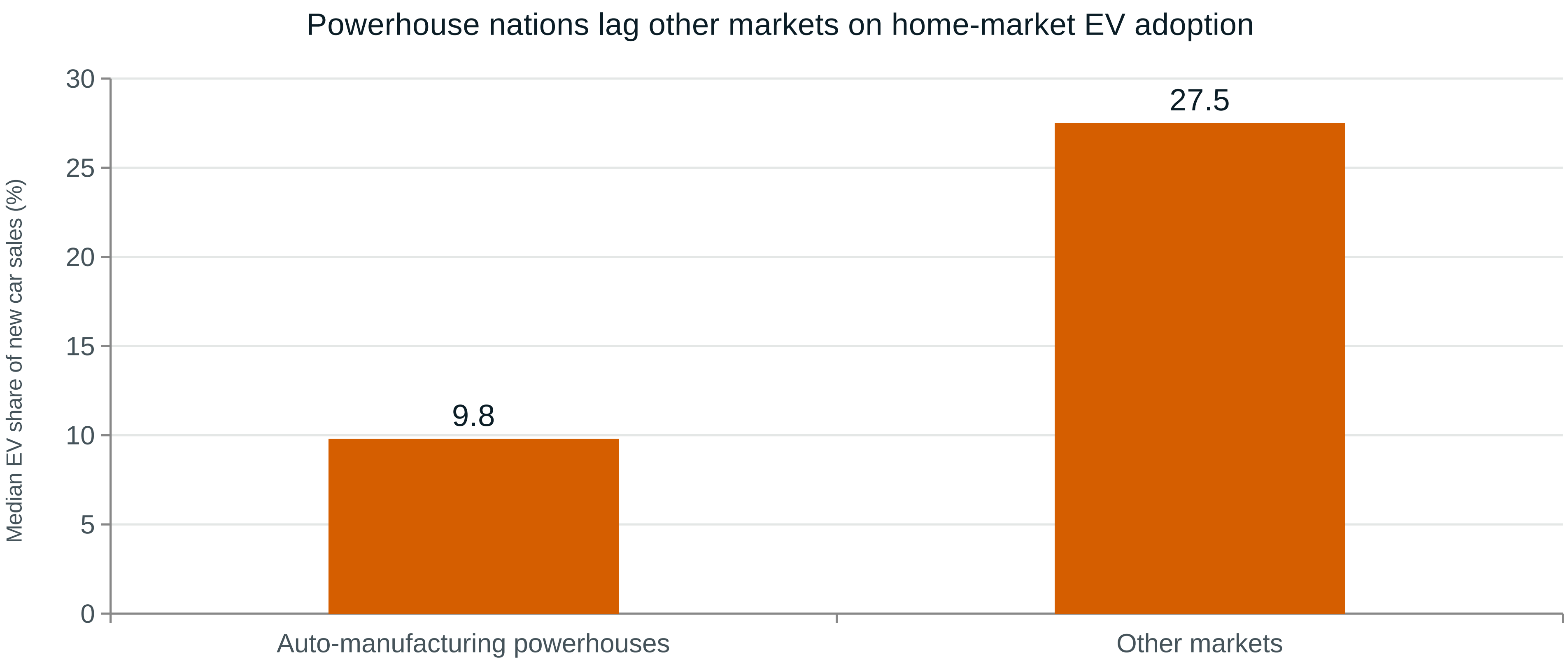
Once a market crosses the 5% tipping point, how fast does it reach the mainstream (50%)?



Takeaway: Only 7 of 60 tracked countries have reached 50% EV sales share so far. China took 6 years to make the same jump Iceland made in 4 -- across roughly 1,000x the sales volume.

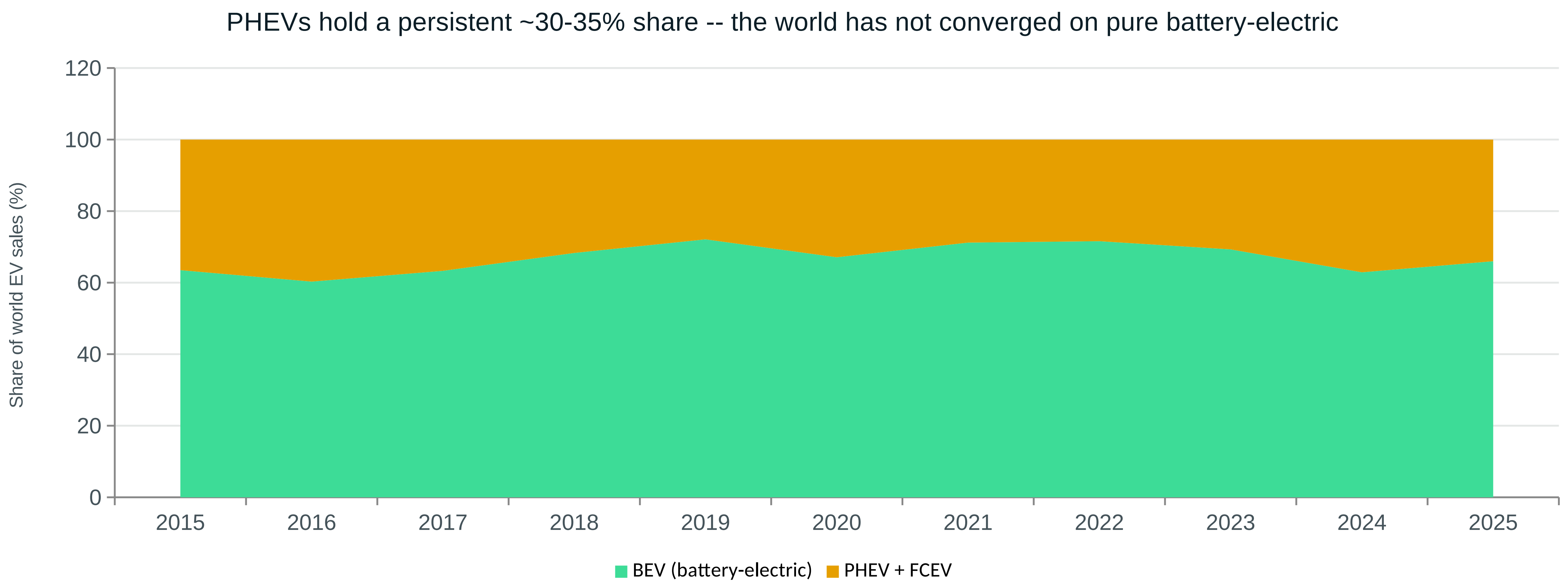
Q4

Are legacy auto-manufacturing powerhouses transforming slower than other markets?



Takeaway: In 2022-2025, the median EV sales share among auto-manufacturing powerhouses was 9.8%, versus 27.5% among other markets -- roughly a three-fold gap.

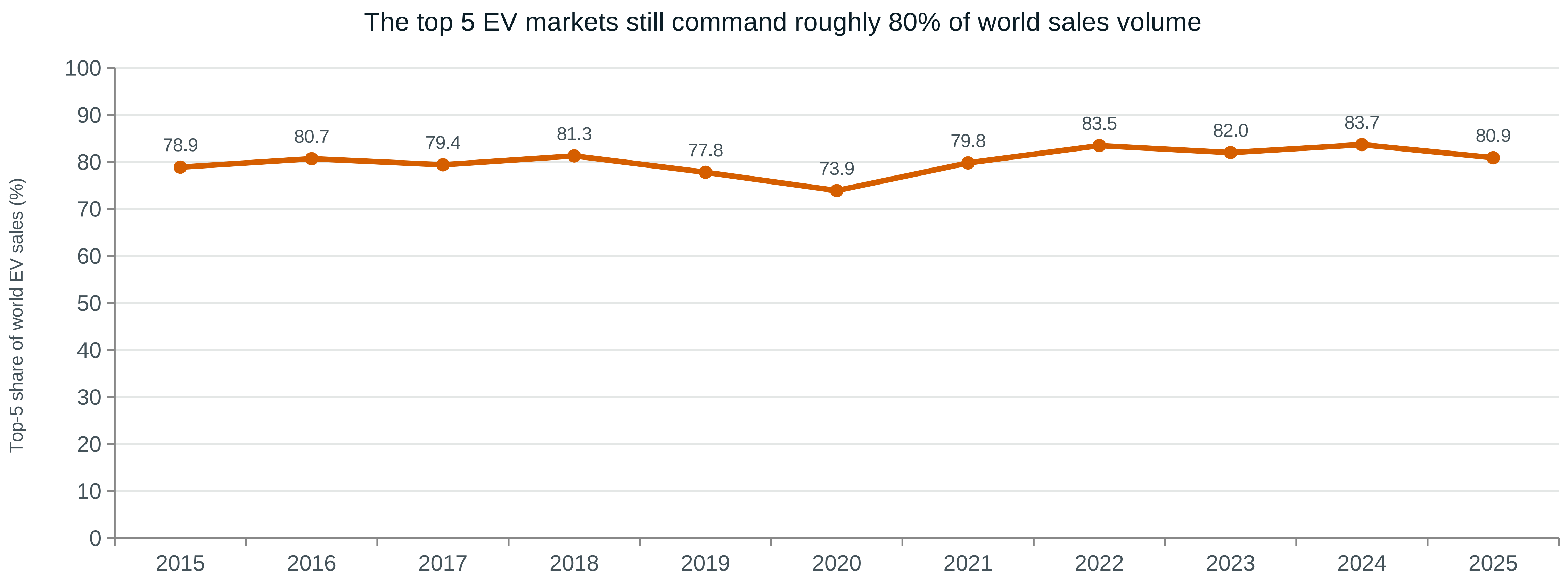
Q5 As the market has scaled, has it settled on pure battery-electric, or stayed hybrid?



Takeaway: Global BEV share has hovered in the 60-65% range through the mid-2020s, after dipping from the pioneer-era 70-90% highs -- plug-in hybrids found a durable foothold rather than disappearing.

Q6

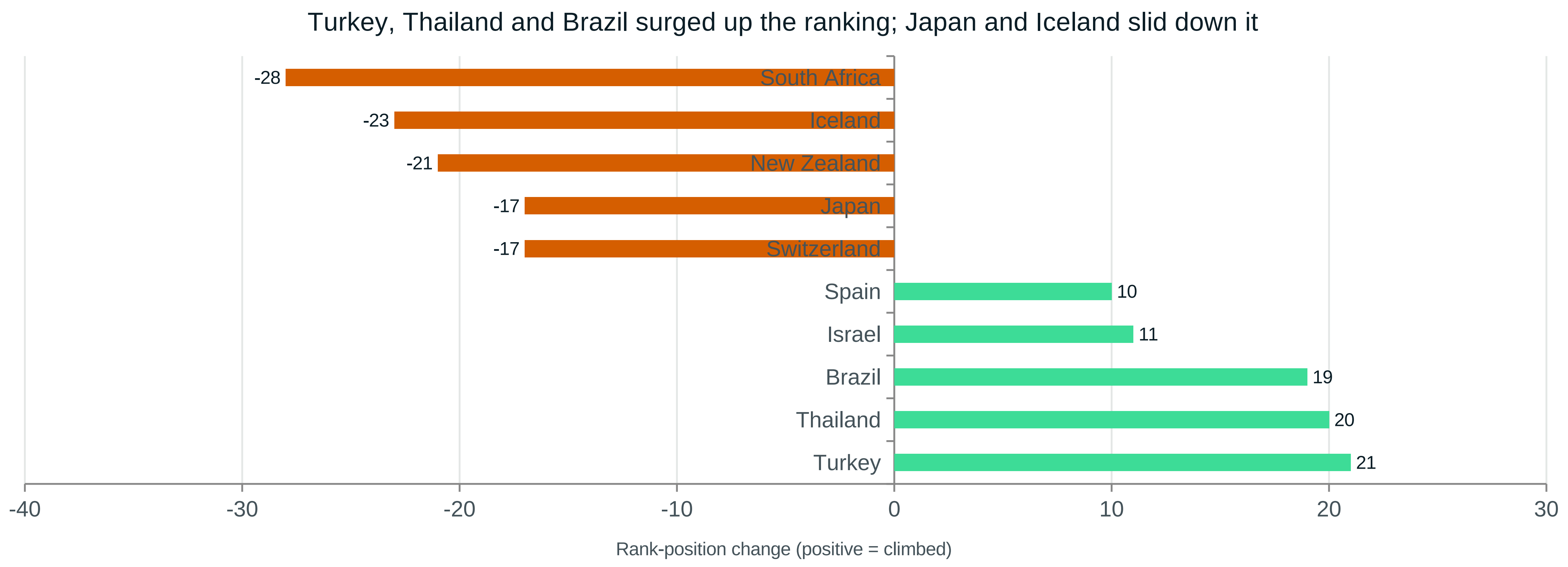
Has the global EV market become more or less concentrated among a few countries?



Takeaway: Concentration held essentially flat -- 78.9% in 2015 (led by China, United States, Norway) to 80.9% in 2025 (led by China, United States, Germany) -- even as dozens of new countries entered the market.

Q7

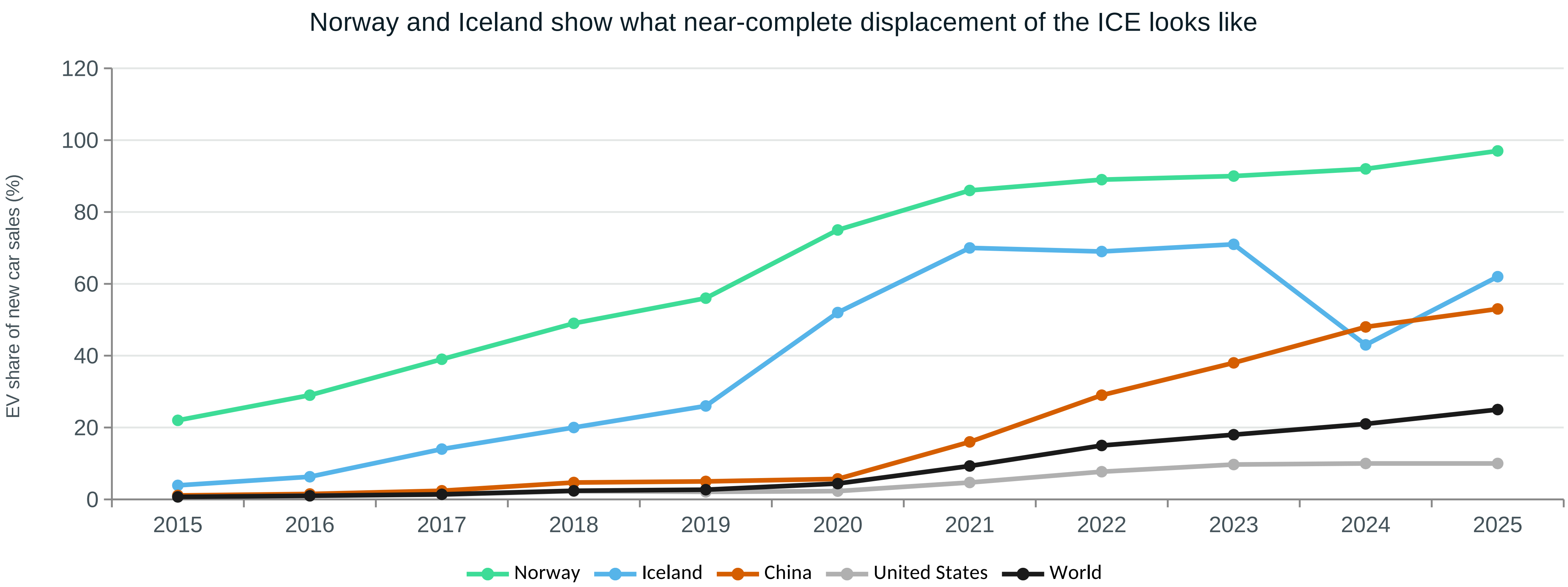
Which countries climbed or fell the most in the global sales-volume ranking, 2015-2025?



Takeaway: Japan went from the world's 5th-largest EV market in 2015 to 22nd in 2025 -- not because its own sales fell, but because dozens of other countries' volumes scaled past its comparatively flat growth.

Q8

What does a near-complete EV transformation actually look like?



Takeaway: Norway reached 97% EV sales share in 2025 and Iceland 62% -- but the World average (25%) and even China (53%) show most of the globe is still mid-transition, not past it.

Explore It Yourself: The Interactive Dashboard

1

Global Overview

World sales/share trend, a country choropleth, and the top-15 markets by volume, for any year and continent filter.

2

Adoption S-Curve Explorer

Spotlight any country against its continent and the world; a bar chart of every market's 5%-to-50% tipping speed.

3

Powerhouses vs. Challengers

Legacy manufacturer vs. other-market comparison, a live growth-vs-maturity scatter explorer, and ranking-churn charts.

Live URL: <your-streamlit-app-url>

Key Takeaways

1. The transformation is real and fast (41x sales growth since 2015) but heavily concentrated: China alone drove 63% of 2025's global volume, and the top 5 markets have held ~80% of sales throughout the decade.

2. Nordic and European markets pioneered the fastest complete transformations (Norway, Iceland: 4-year tipping-to-mainstream sprints), but they are small markets -- China's slower per-capita pace still moves far more metal.

3. Legacy auto-manufacturing powerhouses are electrifying their home markets roughly 3x slower than other countries, a genuine tension between industrial-policy caution and the pace of global change.

4. Most of the world, including the current volume leader China, remains mid-transition (25-53% EV share) rather than past it -- the story of the next decade is still substantially unwritten.

Methodology & Limitations

- Source: IEA Global EV Outlook 2025, republished by Our World in Data under a compatible open license.
- Coverage starts later for markets with no measurable EV sales before roughly 2015-2019 -- those countries simply have fewer rows, not missing data.
- 'Tipping point' (5%) and 'mainstream' (50%) thresholds follow commonly-cited EV-adoption S-curve benchmarks (BloombergNEF, IEA), not a claim inherent to the raw data.
- 'Auto-manufacturing powerhouse' is a fixed list built from OICA global vehicle-production rankings -- an analytical grouping for comparison, not a value judgement.
- The 'Rest of World' aggregate is a residual (World minus all named countries) and can show small negative year-over-year swings from data revisions -- not a literal fleet decline.

Thank You

Global Electric Vehicle Market Analysis, 2010-2025

Repository, dashboard, and notebook links in the README.